Nikhil Mahadevan

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Education

University of Georgia | Athens, GA **Aug 2024 - May 2028**
B.S. in Electrical Engineering, B.S. in Computer Engineering **GPA: 3.93**
Honors: Eagle Scout**,** UGA Dean's List

Technical Skills

Languages: C, C++, ARM Assembly, Python, MATLAB
Embedded Systems: ARM Cortex-M4 (STM32), ARM Cortex-A53 (BCM2837), ATmega328P, ESP32, Zephyr RTOS, NASA F Prime
Hardware & EDA: Altium Designer, KiCad, 4-Layer PCB Design, SPICE, Autodesk Fusion
Tools & Test Equipment: Renode, Git, Oscilloscopes, Logic Analyzers, Multimeters, Soldering

Experience

Flight Software Engineer | Small Satellite Research Laboratory **Sep 2026 - Present**
- Developing CubeSat flight software in C++ and NASA F´ with C Zephyr drivers on STM32 for a ~15-person software team
- Refactored legacy flight state machine and file-recycler code into reviewable C++ modules with documented interfaces
- Implemented a 7-state flight state machine in C++ with voltage debounce, priority arbitration, and safe-shutdown sequencing
- Hardened file-recycler with path abstractions, safer symlink handling, and mutex-based thread safety
- Review teammates' FSW changes; building UART/RS-485 Zephyr drivers and validating them in Renode

Embedded Systems Research Intern | University of Georgia **Aug 2026 - Present**
- Developing C++ drivers across I2C and SPI interfaces to acquire real-time IMU, ambient temperature, and relative humidity telemetry on an ATmega328P-based canine health monitor
- Streaming structured multi-sensor packets to a host application in real time over HC-05 Bluetooth via hardware UART

Electronics Lab Assistant | University of Georgia **Aug 2026 - Present**
- Instructing 100+ students across two labs on analog filters, oscillators, transformers and BJT amplifiers with SPICE verification
- Guiding students with TTL circuit builds and benchtop debugging using oscilloscopes, logic analyzers and multimeters

Engineering Peer Tutor | University of Georgia **Sep 2025 – May 2026**
- Tutored 35+ students across courses mainly focused on Logic Design (CSEE 2220) and Circuit Analysis (ECSE 2170)

Projects

Edge AI Wake-Word Engine (Personal Project) **Aug 2026**
- Engineered register-level C firmware on an STM32L4 (Cortex-M4) for real-time PDM audio acquisition
- Implemented a DC-blocking filter and energy detector in C to isolate speech buffers and reject background noise
- Developed a 921.6 kbps UART streaming driver with SysTick timestamping to transfer raw audio buffers to a host PC
- Deployed an on-device MFCC-based logistic regression model on the Cortex-M4 (<2 KB RAM, sub-millisecond score latency), achieving 92% validation accuracy on a 600-sample balanced dataset

2-Axis Light Tracker (ELEE 4230) **Jun 2026 - Jul 2026**
- Built a battery-powered dual-axis light tracker in bare-metal C on an ATmega328P using MOSFET H-bridges to drive motors
- Integrated an LDR array with Hall-effect and potentiometer feedback for tilt limit protection
- Implemented a fixed-point Kalman filter to smooth ADC readings, reducing peak noise from 3 LSBs to < 1 LSB without using a FPU
- Engineered a register-level 115.2 kbps UART driver to stream real-time Kalman-filtered tracking telemetry
- Designed a 4-layer PCB in KiCad with dedicated ground planes to isolate motor switching noise from analog sensor traces

Multifunction Instrumentation System (ECSE 2920) **Jan 2026 - Apr 2026**
- Built a Raspberry Pi digital multimeter and function generator in Python with a ±12V split-rail supply and active virtual ground
- Designed a discrete SAR ADC via SPI digipot and comparator for a 0-5V voltmeter, 500-10 kΩ ohmmeter, and ±5V reference
- Built function generators (0-10 Vpk, 0-10 kHz) and a 0-10 kHz frequency counter with op-amp conditioning
- Improved DC resolution 6x and voltmeter precision 2x via digipot calibration in software
- Designed a 4-layer mixed-signal Pi HAT PCB in KiCad with dedicated power and ground planes